40. Prove or disprove that if you have an 8-gallon jug of water and two empty jugs with capacities of 5 gallons and 3 gallons, respectively, then you can measure 4 gallons by successively pouring some or all of the water in a jug into another jug.

Solution: Let, we have 3 jugs J1, J2 and J3 respectively. J1 has a capacity of 8 gallons, J2 has a capacity of 5 gallons and J3 has a capacity of 3 gallons. We shall outline a step-by-step procedure on how to measure 4 gallons.

	J1	J2	J3
Step 1:	8	0	0
Step 2:	3	5	0
Step 3:	3	2	3
Step 4:	6	2	0
Step 5:	6	0	2
Step 6:	1	5	2
Step 7:	1	4	3

$\therefore$ We finally measure 1 gallon in jug J1, 4 gallon in jug J2 and 3 gallon in jug J3. □

41. Verify the $(3x+1)$ conjecture for these integers.

a) 6 b) 7 c) 17 d) 21

Soln: a) $T(6)=6/2=3$, $T(3)=3\cdot3+1=10$ $T(10)=10/2=5$ $T(5)=5\cdot3+1=16$ $T(16)=16/2=8$ $T(8)=8/2=4$ $T(4)=4/2=2$ $T(2)=2/2=1$

b) $T(7)=7\cdot3+1=22$ $T(22)=22/2=11$ $T(11)=11\cdot3+1=34$ $T(34)=34/2=17$ $T(17)=17\cdot3+1=52$ $T(52)=52/2=26$ $T(26)=26/2=13$ $T(13)=13\cdot3+1=40$ $T(40)=40/2=20$ $T(20)=20/2=10$ $T(10)=10/2=5$ $T(5)=5\cdot3+1=16$ $T(16)=16/2=8$ $T(8)=8/2=4$ $T(4)=4/2=2$ $T(2)=2/2=1$

c) $T(17)=17\cdot3+1=52$ $T(52)=52/2=26$ $T(26)=26/2=13$ $T(13)=13\cdot3+1=40$ $T(40)=40/2=20$ $T(20)=20/2=10$ $T(10)=10/2=5$ $T(5)=5\cdot3+1=16$ $T(16)=16/2=8$ $T(8)=8/2=4$ $T(4)=4/2=2$